MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGG			
END SEMESTER EXAMINATION		**JULY – NOV, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Automobile Engg**	**Course Type**	**PE**
Course Name	**Design of IC Engine**	**Course Code**	**AAE 4035**
Semester	**VII**	**Max Marks**	**50**
Date	**27 – 11 - 2024**	**Duration**	**03 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	The cylinder of a four-stroke diesel engine has the following specifi cations: Brake power = 8.5 kW, Speed = 1500 rpm, Indicated mean effective pressure = 0.4 MPa, Mechanical efficiency = 80%, Maximum gas pressure = 4.5 MPa, The cylinder liner and head are made of grey cast iron FG 260 (σ_{ut} = 260 N/mm^2 and μ =0.25). The studs are made of plain-carbon steel 40C8 (σ_{yt} = 380 N/mm^2). The factor of safety for all parts is 5. Assume *l*/D = 1.5, K = 0.162, Calculate: (i) bore and length of the cylinder liner (ii) thickness of the cylinder liner	04	1	1, 2, 3, 6, 10,12	1	C1, C2, C7	3
1B	From the question 1A data, calculate (iii) Circumferential Stress (iv) Longitudinal Stress (v) Net stresses	03	1	1, 2, 3, 6, 10,12	1	C1, C2, C7	3
1C	From the question 1A data, calculate (vi) Thickess of cylinder head (vii) size, number and pitch of studs & nominal diameter of studs	03	1	1, 2, 3, 6, 10,12	1	C1, C2, C7	3
2A	The following data is given for a four-stroke diesel engine: Cylinder bore = 120 mm, Length of stroke = 180 mm, Speed = 2000 rpm, Brake mean effective pressure = 0.75 MPa, Maximum gas pressure = 4 MPa, Fuel consumption = 0.3 kg per	03	2	1, 2, 3, 6, 10,12	2	C1, C2, C7	3

	BP per hr, Higher calorific value of fuel = 42,000 kJ/kg, Assume that 5% of the total heat developed in the cylinder is transmitted by the piston. The piston is made of grey cast iron and the permissible tensile stress is 37.5 N/mm^2 (k = 46.6 W/m/°C). The temperature difference between the centre and edge of the piston head is 220°C. Mechanical efficiency = 80 %. (i) Calculate the thickness of the piston head by strength consideration. (ii) Calculate the thickness of the piston head by thermal consideration. (iii) State whether the ribs are required.						
2B	From the question 2A data, and using the assumptions, Radial pressure on the side wall = 0.035 MPa. Permissible tensile stress for piston rings = 90 N/mm^2 (iv) calculate the number and thickness of piston ribs. (v) State whether a cup is required in the top of piston head.(vi) If so, calculate the radius of the cup, (vii) radial width of the piston rings; (viii) axial thickness of the piston rings.	03	2	1, 2, 3, 6, 10,12	2	C1, C2, C7	3
2C	From the question 2A data, and assumption, allowable bearing pressure = 0.45 MPa, μ=0.1, calculate (x) gap between the free ends of the piston ring before assembly; (xi) gap between the free ends of the piston ring after assembly; (xii) width of the top land; (xiii) width of the ring grooves; (xiv) thickness of the piston barrel; (xv) length of piston skirt; (xvi) length of piston	04	2	1, 2, 3, 6, 10,12	2	C1, C2, C7	3
3A	The following data is given for the connecting rod of a diesel engine: Cylinder bore = 100 mm, Length of connecting rod = 360 mm, Maximum gas pressure = 3 MPa, Factor of safety against buckling failure = 5, Allowable bearing pressure for piston pin bearing = 13 MPa, Allowable bearing pressure for crank pin bearing = 11 MPa, Length of stroke = 150 mm, Mass of reciprocating parts = 1.7 kg, Engine speed = 1900 rpm, Thickness of bearing bush = 3 mm, Material of cap = steel 40C8, Yield strength of cap material = 380 N/mm^2 Factor of safety for cap = 4, Material of bolts = chromium molybdenum steel, Yield strength of bolt material = 450 N/mm^2 , Factor of safety for bolts = 5, Density of connecting rod = 7800 kg/m^3 Assume, (l/d) ratio for piston pin bearing = 1.5, (l/d) ratio for crank pin bearing = 1.25 Calculate: (i) dimensions of the cross-section of connecting rod; (ii) dimensions of small and big end bearings	04	3	1, 2, 3, 6, 10,12	3	C1, C2, C7	3

3B	From the data in question 3A, calculate (iii) nominal diameter of bolts for the cap; (iv) thickness of cap; and (v) magnitude of whipping stress	04	3	1, 2, 3, 6, 10,12	3	C1, C2, C7	3
3C	Explain the design considerations with respect to length of the connecting rod.	02	3	1, 2, 3, 6, 10,12	3	C1, C2, C7	2
4A	Design a centre crankshaft for a ingle-cylinder vertical engine using the following data: Cylinder bore = 130 mm, (L/r) ratio = 4.5, Maximum gas pressure = 3.5 MPa, Length of stroke = 170 mm Weight of flywheel cum belt pulley = 1.5 kN, Total belt pull = 2.5 kN, Width of hub for flywheel cum belt pulley = 200 mm, The torque on the crankshaft is maximum when the crank turns through 25° from the top dead centre and at this position the gas pressure inside the cylinder is 2 MPa. The belts are in the horizontal direction.Allowable bending stress for the crank pin is 75 N/mm^2 (i) Calculate the bearing reactions, the dimensions and stress in the crank pin (ii) Calculate the dimensions and stress in left hand crank web	04	3	1, 2, 3, 6, 10,12	3	C1, C2, C7	3
4B	From the data in question 4A, calculate (iii) the dimensions of the shaft and stress in the shaft under flywheel	03	3	1, 2, 3, 6, 10,12	3	C1, C2, C7	3
4C	With a neat sketch, classify the types of crank shaft used in IC engines.	03	3	1, 2, 3, 6, 10,12	3	C1, C2, C7	2
5A	Design a rocker arm for the exhaust valve of a four-stroke engine using the following data: Effective length of each arm = 200 mm, Angle between two arms = 135°, Diameter of valve head = 80 mm, Lift of valve = 30 mm, Mass of valve = 0.6 kg, Engine speed = 800 rpm, Back pressure when the exhaust valve opens = 0.5 MPa, Maximum suction pressure = 0.03 MPa below Atmosphere. Calculate (i) the total force on the rocker arm considering an acceleration of 150 m/s^2. Assume permissible bending stress = 70 N/mm^2, permissible shear stress = 42 N/mm^2, permissible compressive stress = 50 N/mm^2, permissible bearing pressure = 5 MPa. (ii) Reactions and dimensions of the fulcrum pin (iii) Stress in the fulcrum pin	04	4	1, 2, 3, 6, 10,12	3	C1, C2, C7	3
5B	From the question 5A, calculate (iv) Dimensions of roller pin (v) Shear stress & Bending stresses at the roller pin	03	4	1, 2, 3, 6, 10,12	3	C1, C2, C7	3

5C	From the data question 5A, calculate (vi) Thickness and proportions of cross secton for the rocker arm (vii) core and nominal diameter of the tappet.	03	4	1, 2, 3, 6, 10,12	3	C1, C2, C7	3

Retain only one. Delete the others

**** Applicable for only IET-accredited courses (modules). The IET columns may be deleted for non-IET courses.***

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	10
2	10
3	20
4	10

BL	Total Marks
2	**05**
3	**45**
4	**-**
5	**-**